

Su-won Jin

Introduction & Projects



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Introduction



✉ hmnc0906@naver.com

🐙 <https://github.com/Jinsuwon>

“I want to study robotic intelligence that understands human language and interacts with the real world”

Name

- Su-won Jin (진수원)

Education

- Korea Maritime and Ocean University,
- B.S. in Electrical and Electronic Engineering
- Total GPA: 4.0 / 4.5, Major GPA: 3.98 / 4.5

Experience

- Undergraduate Researcher / Jul. 2024 – Jan. 2025
- Embedded Signal Processing Lab
- Course-based Intern / Jan. 2025 – Feb. 2025
- SEANET, Engineering Design Team

Research Interests

- Vision-Language-Model
- 3D Scene Understanding
- Robot Perception
- Robot Task Planning

A BLE-Based Bus Boarding System for the Visually Impaired

A BLE-Based Bus Boarding System for the Visually Impaired

- **Barriers to Bus Use Among the Visually Impaired**

Reason	Percentage (%)
Difficulty identifying the desired bus number	24.5
Difficulty locating boarding and exit doors	8.4

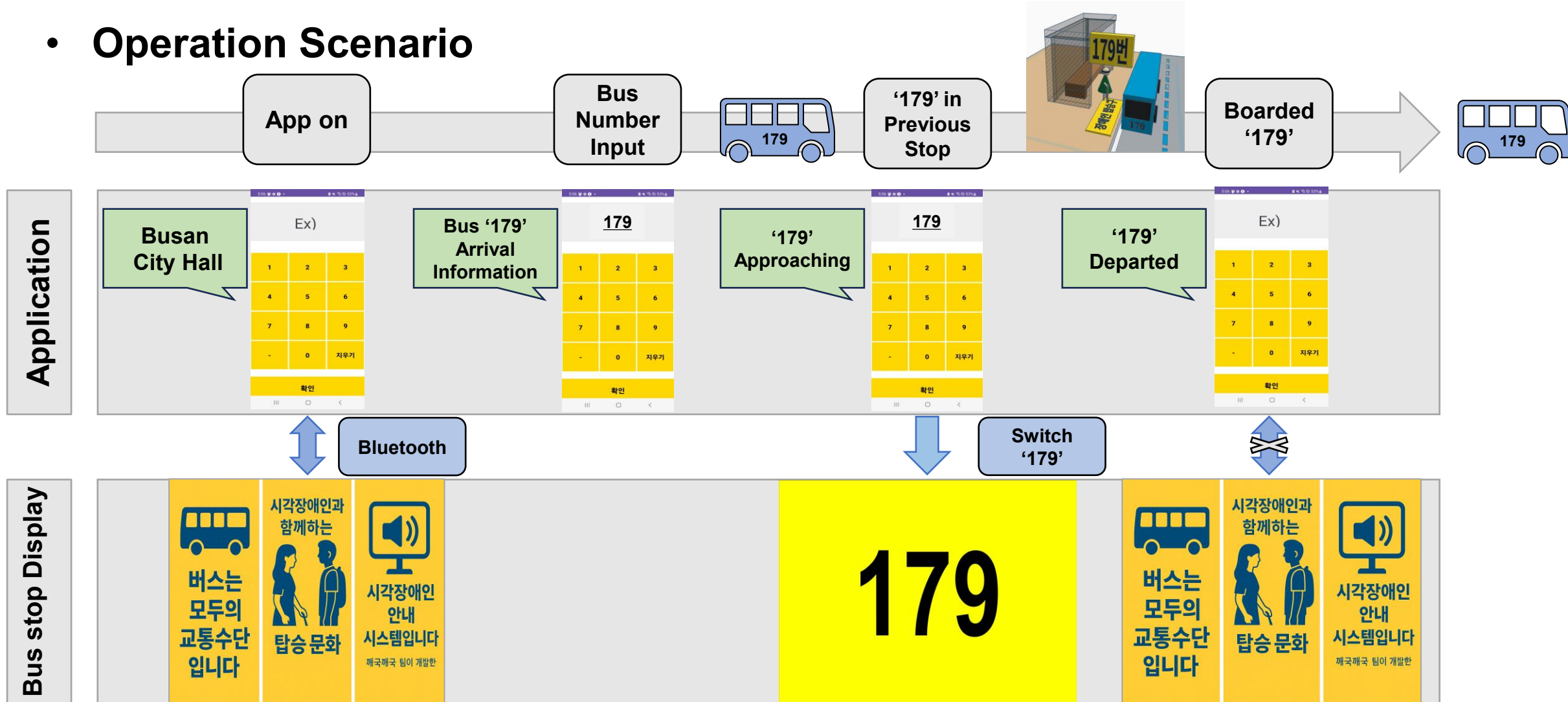
*Source: Wello, Korea Blind Union

- **Solutions**

- Voice-guided application for bus number identification
- Bus stop display with a designated boarding zone

A BLE-Based Bus Boarding System for the Visually Impaired

- Operation Scenario



A BLE-Based Bus Boarding System for the Visually Impaired

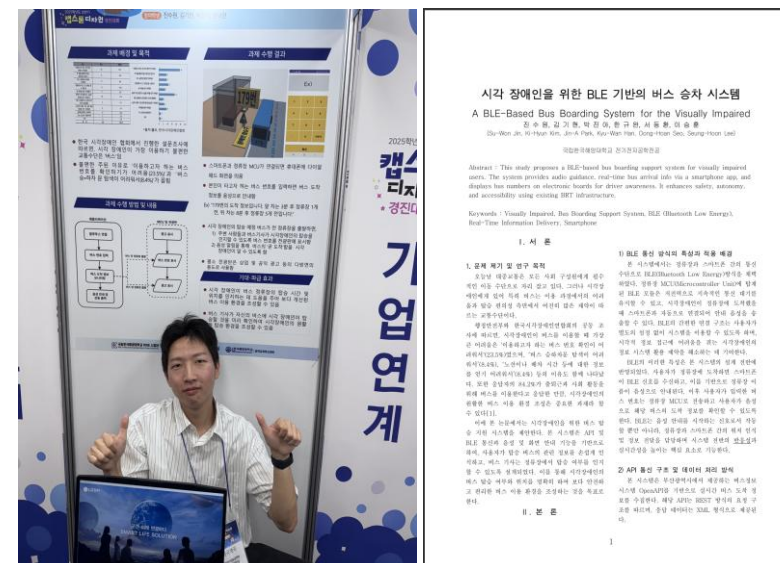
• Results

2025.03 ~ 2025.06

- BLE-based app-display bus boarding system
- Competition & Conference participation
- Gold prize in departmental competition

• My Contribution

- OpenAPI integration & XML parsing
- Event-driven display/alert logic
- Team Leader
- Paper preparation & submission
- Poster presentations at competitions and a conference



< Competition >

< Paper >

Real-time Toilet Paper Remaining Level Monitoring System

Real-time Toilet Paper Remaining Level Monitoring System

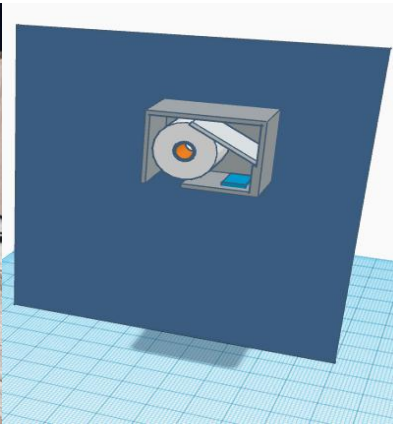
- **Problems**

- Toilet paper availability is critical in high-traffic public restrooms
- Manual inspection makes efficient restroom management difficult in multi-stall environments

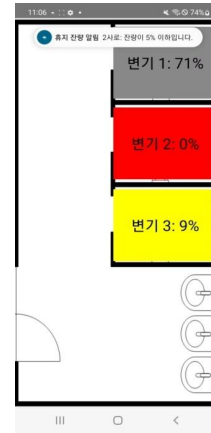
- **Solutions**



< Dispenser >



< Installation Concept >

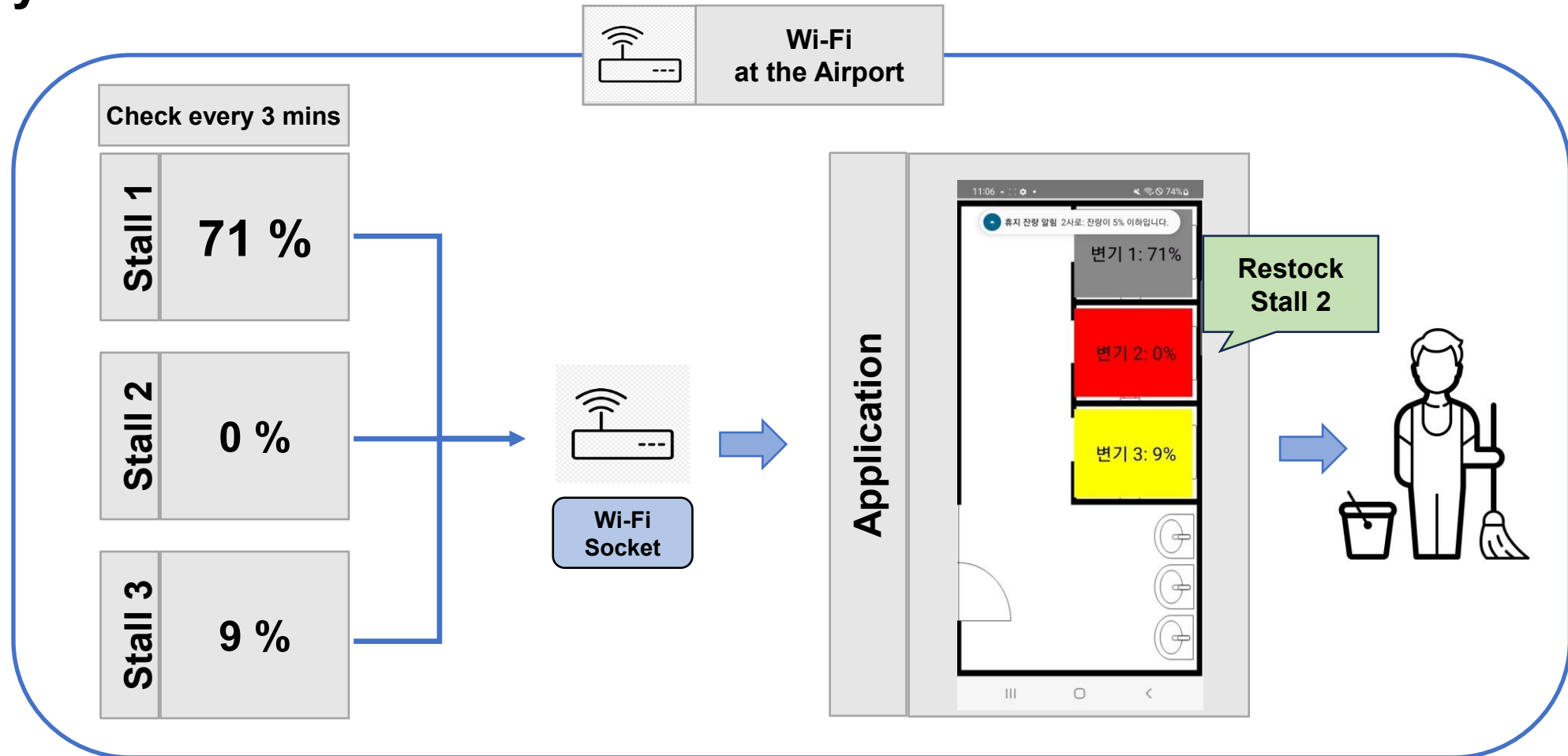


< Application >

- Arduino-based smart toilet paper dispenser for level detection
- Wi-Fi-based monitoring and management application

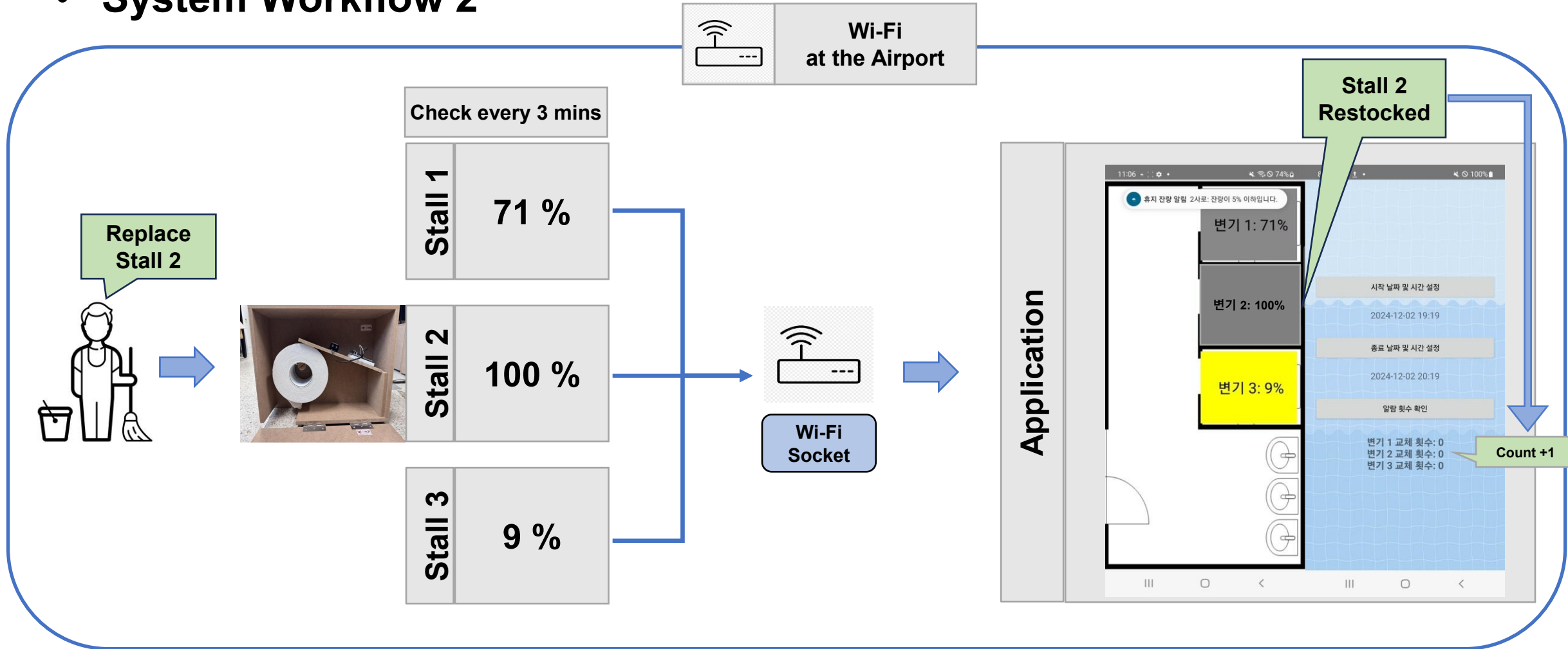
Real-time Toilet Paper Remaining Level Monitoring System

- System Workflow 1



Real-time Toilet Paper Remaining Level Monitoring System

- System Workflow 2



Real-time Toilet Paper Remaining Level Monitoring System

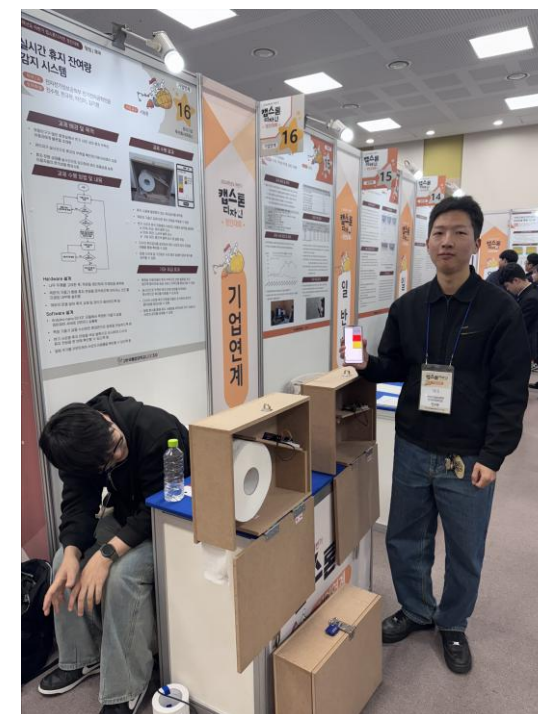
- **Results**

2024.09 ~ 2024.12

- IMU-based toilet paper level detection dispenser
- Real-time monitoring application
- Competition awards & Patent pending

- **My Contribution**

- Application UI development & sensor data visualization
- IMU sensor data processing logic
- Replacement event detection
- Team Leader
- Poster presentations at competitions

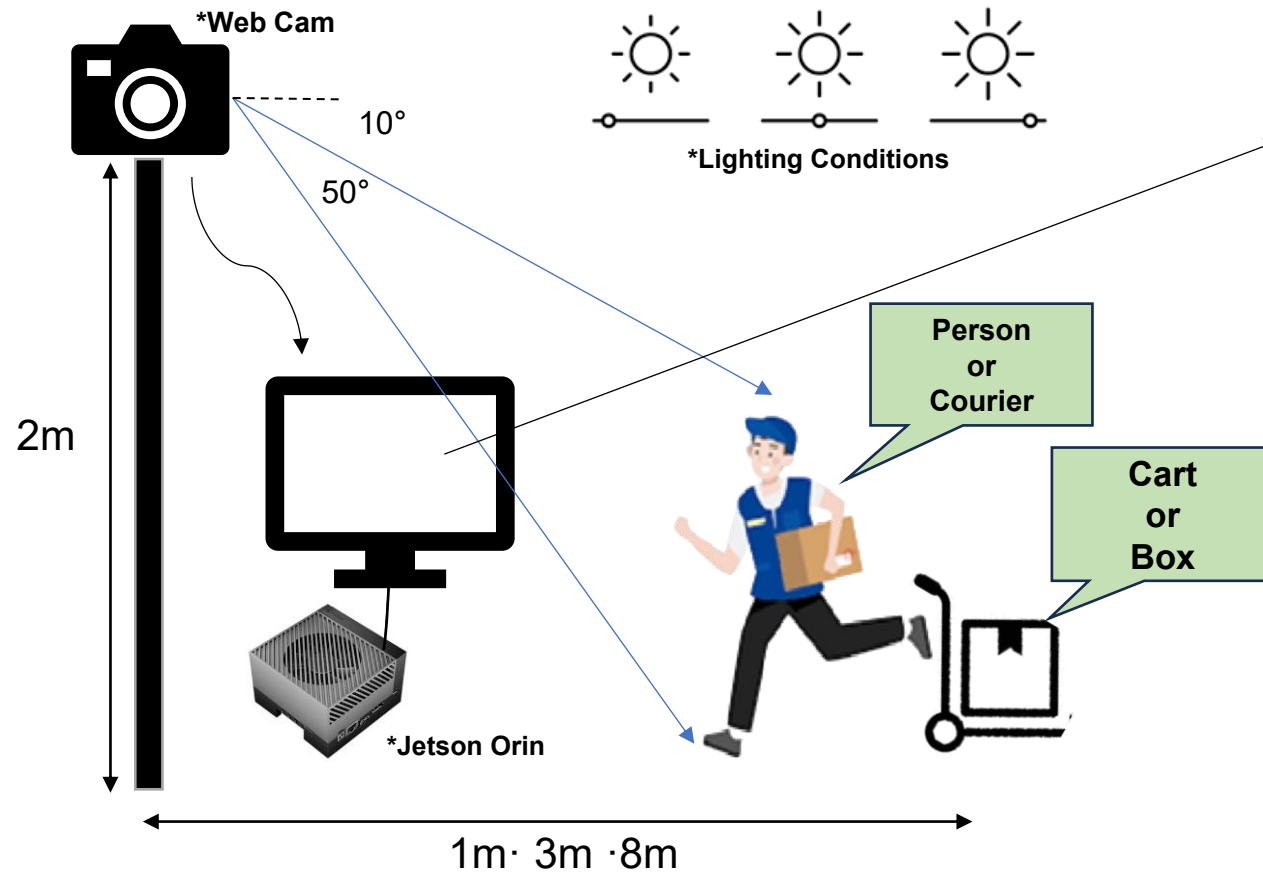


< Competition >

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

- **Experimental Overview:** Development of an object recognition system for an unmanned parcel locker



[timestamp] A person is approaching
[timestamp] A courier is approaching with
(cart or box)
⋮



< Experimental Setup >

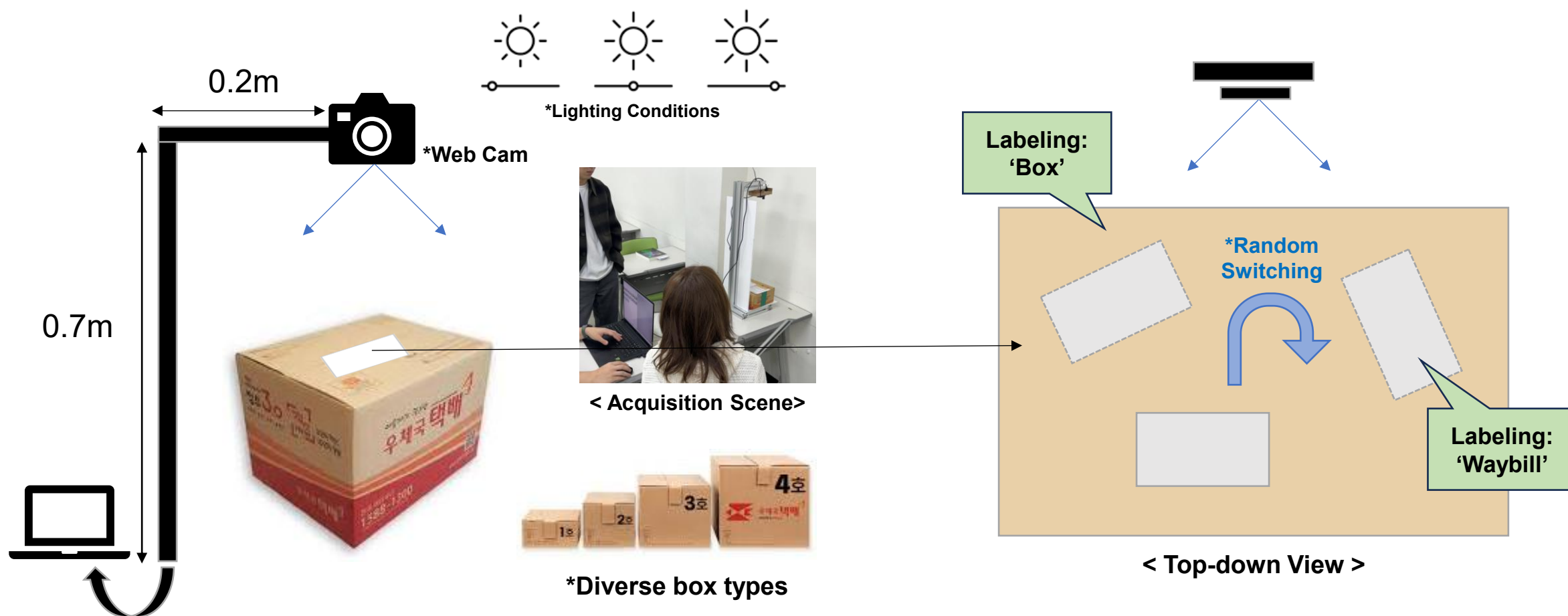


< Target Object >

< YOLO-based target object classification and predefined message output >

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

- Dataset Acquisition & Labeling



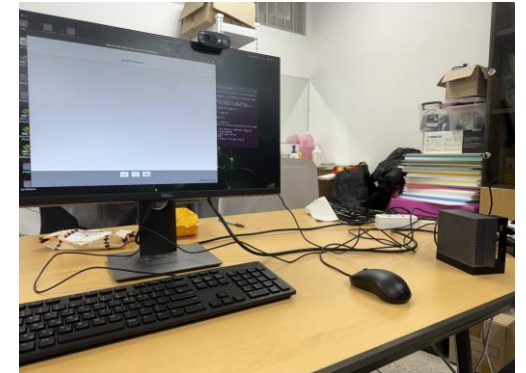
< Dataset with diverse lighting, box types, and waybill positions >

Development of High-Density Smart Parcel Terminal Storage Infrastructure and Management Technology

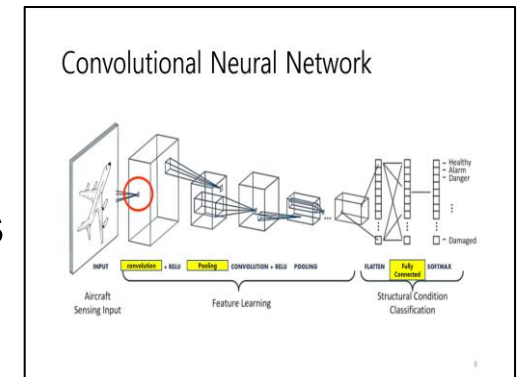
• My Contribution

2024.07 ~ 2024.12

- **Testbed construction for object recognition model execution**
 - UI, Initial settings, Jetson Orin deployment
- **Validation environment design**
 - camera height, lighting, clothing, angle, and object conditions
- **Top-down dataset acquisition & annotation**
 - diverse lighting, box, and waybill positions
- **Fundamental study of neural networks and evaluation metrics**
 - Seminar presentation on CNN/RNN models and captioning metrics



< Jetson Orin Deployment >



< Seminar PPT >

< Click to view on [GitHub](#) >

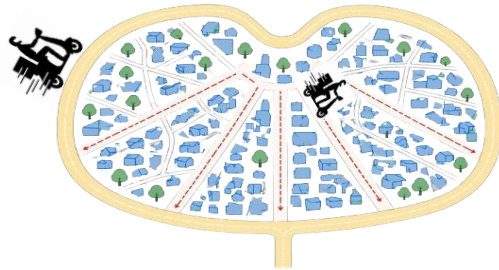
Inter – University B.SORI Project Competition

Inter –University B.SORI Project Competition

- **Background**

- Inter-university idea competition on local community issues
- Two-day team formation and idea generation
- Ten-day idea refinement and PPT presentation

- **Objective: Motorcycle noise reduction for different residential environments**



< Hillside Residential Area >

- **Paved detour road installation**

- Uphill motorcycle noise in hillside area
- Detour-based noise reduction strategy



< Urban Area >

- **Rear-view noise enforcement camera**

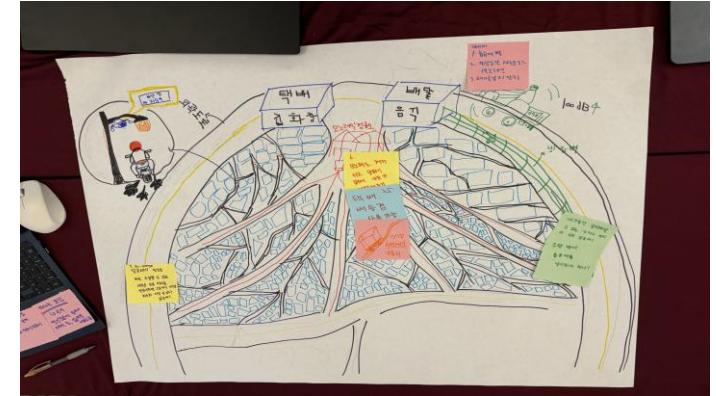
- Enhanced nighttime noise enforcement

Inter –University B.SORI Project Competition

- **Results**

- Local issue analysis and solution ideation
- Noise reduction ideas for residential environments
- Concept proposal PPT
- Encouragement award

2024.09 ~ 2024.09



< Initial Concept Sketch >

- **My Contribution**

- Project direction and team coordination
- Noise issue and regional characteristics analysis
- Team Leader
- Online PPT presentation



< Concept Proposal PPT >

< Click to view on [GitHub](#) >

Proposal for a Non-Contact Surge(Lightning) Counter

Proposal for a Non-Contact Surge(Lightning) Counter

- **Background**

- Student railway idea competition hosted by the Korean Society for Railway
- Team-based railway idea proposal in PPT format
- Participation in a non-contact surge (lightning) counter proposal under faculty guidance

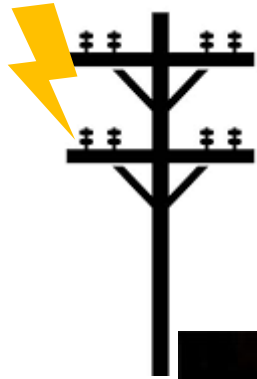
- **Objective**

- Proposal of a non-contact surge counter using electromagnetic induction without external power
- Improvement of installation complexity, external power dependency, and high cost in conventional equipment

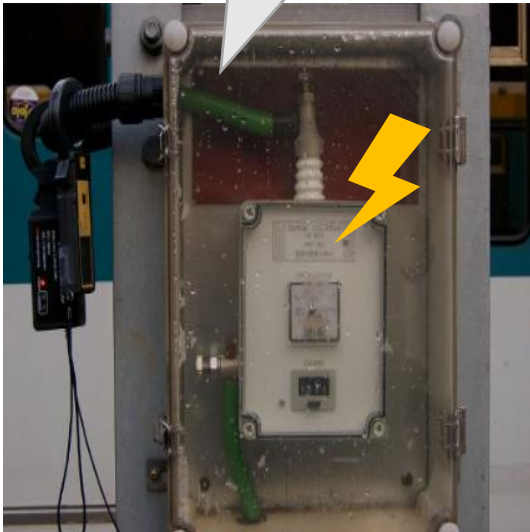


< Proposal Counter >

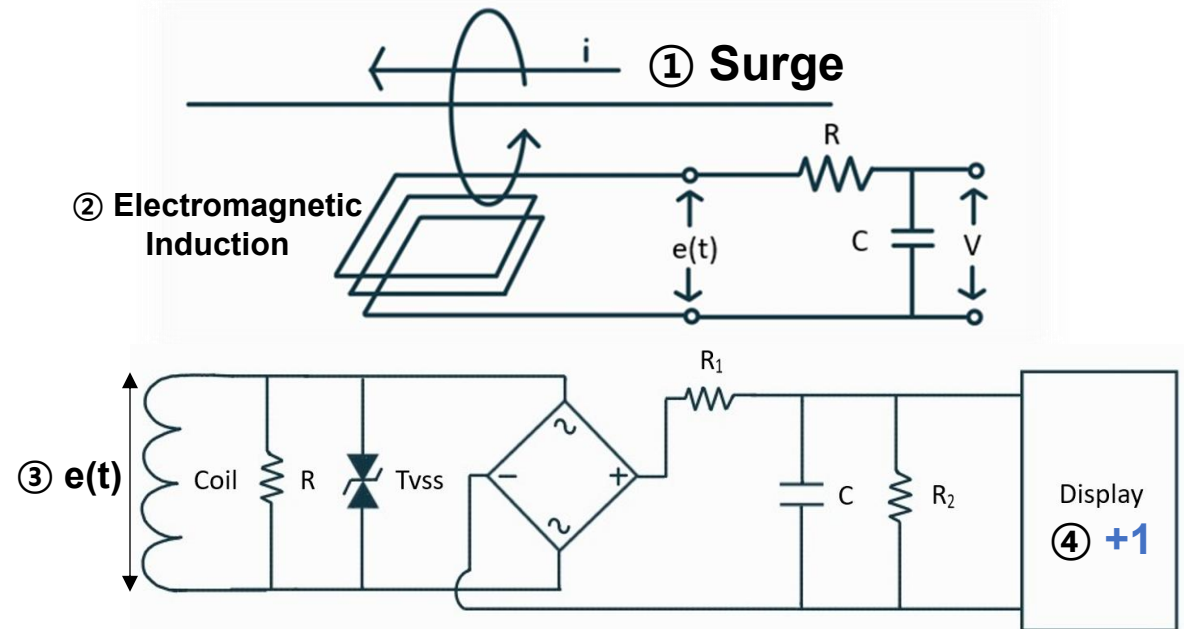
Proposal for a Non-Contact Surge(Lightning) Counter



- *Installation complexity
- *External power dependency
- *High cost



< Conventional Counter >



*Circuit Diagram



A Non-Contact Surge Counter

*arrester down conductor

< Installation Example >

Proposal for a Non-Contact Surge(Lightning) Counter

2024.05 ~ 2024.06

- **Results**

- Non-contact surge counter concept proposal
- Improvement measures over conventional equipment
- Concept proposal PPT



< Concept Proposal PPT >

< Click to view on [GitHub](#) >

- **My Contribution**

- Study of surge counters, surge, electromagnetic induction, and circuit configurations
- Analysis of conventional surge counter structures and conceptualization of a non-contact approach
- Team Leader
- PPT preparation and incorporation of faculty feedback